

Further Statistics 1

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Chapter 4: Hypothesis Testing for Poisson and Binomial Parameters

Why this chapter matters

You already know how to carry out a hypothesis test for a binomial proportion using one-tailed critical regions — this chapter extends that same machinery to the Poisson parameter λ , and generalises it to two-tailed tests for both distributions. The logic doesn't change; what's new is finding critical regions on both tails at once, and applying the whole method to a distribution you met only in Chapter 2.

Concept: the hypothesis testing framework (recap and extension)

Every hypothesis test in this chapter follows the same structure:

1. Define the parameter of interest (p for Binomial, λ for Poisson) and state H_0 and H_1 .
2. Decide whether the test is one-tailed ($H_1: p < p_0$ or $p > p_0$) or two-tailed ($H_1: p \neq p_0$).
3. Find the critical region(s): the set of values of the test statistic that would lead you to reject H_0 , using the given significance level.
4. Compare the observed value of the test statistic to the critical region and state a conclusion in context.

The new content is: (a) applying this to Poisson data, and (b) splitting the significance level across two tails.

Key vocabulary

Term	Meaning
Null hypothesis, H_0	The “no change” statement about the parameter, assumed true unless evidence suggests otherwise
Alternative hypothesis, H_1	The claim being tested for, e.g. the parameter has increased/decreased/changed
Significance level	The threshold probability (e.g. 5%) for rejecting H_0 when it is actually true
Critical region	The set of values of the test statistic that lead to rejection of H_0
Critical value	The boundary value(s) of the critical region
Actual significance level	The true probability of a Type I error once the critical region is fixed on discrete data (usually not exactly equal to the nominal level, e.g. 5%, because of discreteness)

A common misreading to avoid: the significance level is $P(\text{we reject } H_0 \mid H_0 \text{ is actually true})$ — it is a statement about how the *test* behaves, not the probability that H_0 itself is true. “Significant at the 5% level” does not mean “there’s only a 5% chance H_0 is correct”; it means that *if* H_0 were true, results this extreme would only turn up 5% of the time by chance, which is why they count as good evidence against H_0 . This distinction is worth stating explicitly, because it is exactly the definition of a Type I error, formalised properly in Chapter 8.

Procedure: finding a one-tailed critical region (Poisson mean or Binomial proportion)

1. State $H_0: \lambda = \lambda_0$ and $H_1: \lambda < \lambda_0$ (or $\lambda > \lambda_0$).
2. Under H_0 , $X \sim \text{Po}(\lambda_0)$.
3. For a lower-tail test, find the largest c such that $P(X \leq c) \leq \text{significance level}$. The critical region is $X \leq c$.
4. For an upper-tail test, find the smallest c such that $P(X \geq c) \leq \text{significance level}$. The critical region is $X \geq c$.
5. Compare the observed value to the critical region and conclude in context.

Worked example 1: one-tailed Poisson test

EXAMPLE 4.1

Question. A call centre receives calls at a mean rate of 6 per 10-minute period. After a new advertising campaign, a random 10-minute period is monitored and 11 calls are received. Test, at the 5% significance level, whether there is evidence that the mean rate of calls has increased.

Step 1. Let X be the number of calls in a 10-minute period. Under H_0 , $X \sim \text{Po}(6)$. $H_0: \lambda = 6$, $H_1: \lambda > 6$. **(B1)**

Step 2. We need the smallest c such that $P(X \geq c) \leq 0.05$, i.e. $P(X \leq c-1) \geq 0.95$, using $\text{Po}(6)$ cumulative probabilities. From tables, $P(X \leq 9) = 0.9161$ and $P(X \leq 10) = 0.9574$. **(M1)**

Step 3. So $P(X \geq 10) = 1 - 0.9161 = 0.0839$ (too big) and $P(X \geq 11) = 1 - 0.9574 = 0.0426 (\leq 0.05)$. The critical region is $X \geq 11$, with actual significance level 4.26%. **(A1)**

Step 4. The observed value 11 lies in the critical region. **(M1)**

Answer: There is sufficient evidence, at the 5% level, to conclude that the mean rate of calls has increased. **(A1)**

EXAM TIP

It is common to write the critical region the wrong way round (e.g. $X \leq 11$ for an upper-tail test). Always sanity-check: “does my critical region contain values that support H_1 ?” For $H_1: \lambda > 6$, large values of X support H_1 , so the critical region must be an upper tail like $X \geq 11$. This one check catches most sign errors before they cost marks.

Concept: two-tailed tests and splitting the significance level

For a two-tailed test ($H_1: \text{parameter} \neq \text{value}$), the significance level is split between both tails — usually equally. A 5% two-tailed test uses 2.5% in each tail; a 10% two-tailed test uses 5% in each tail. You find a lower critical region and an upper critical region separately, exactly as in the one-tailed procedure, but using half the significance level each time.

Worked example 2: two-tailed test for a binomial proportion
EXAMPLE 4.2

Question. A machine is supposed to produce components with a defect rate of 0.15. In a random sample of 20 components, 1 is found to be defective. Test, at the 10% significance level, whether there is evidence that the defect rate is different from 0.15.

Step 1. Let X be the number of defectives in the sample of 20. Under H_0 , $X \sim B(20, 0.15)$. $H_0: p = 0.15$, $H_1: p \neq 0.15$. Since this is two-tailed, use 5% in each tail. **(B1)**

Step 2. Lower tail: find the largest c such that $P(X \leq c) \leq 0.05$. From $B(20, 0.15)$ tables, $P(X \leq 0) = 0.0388$ and $P(X \leq 1) = 0.1756$. So the lower critical region is $X \leq 0$. **(M1A1)**

Step 3. Upper tail: find the smallest c such that $P(X \geq c) \leq 0.05$, i.e. using $P(X \leq c-1) \geq 0.95$. From tables, $P(X \leq 6) = 0.9327$ and $P(X \leq 7) = 0.9781$, so $P(X \geq 7) = 1 - 0.9327 = 0.0673$ and $P(X \geq 8) = 1 - 0.9781 = 0.0219$. The upper critical region is $X \geq 8$. **(M1A1)**

Step 4. The critical region is $X \leq 0$ or $X \geq 8$. The observed value $X = 1$ does not lie in either critical region. **(M1)**

Answer: There is insufficient evidence, at the 10% level, to conclude that the defect rate is different from 0.15. **(A1)**

EXAM TIP

The most common two-tailed error is halving the wrong quantity, or forgetting to halve at all and hunting for a critical region using the full 10%. Write “significance level per tail = $10\%/2 = 5\%$ ” as an explicit written line every single time — it is a one-mark habit that prevents a much larger cascade of errors.

Concept: interpreting a p-value approach

An equivalent (and sometimes faster) method is to calculate the actual probability of the observed value or a more extreme value occurring under H_0 (the p-value), and compare it directly to the significance level (or half of it, for a two-tailed test), rather than finding the critical region explicitly first.

Worked example 3: p-value approach
EXAMPLE 4.3

Question. Repeat Example 1 (call centre, $X \sim \text{Po}(6)$, observed value 11, $H_1: \lambda > 6$) using a direct probability comparison instead of finding the critical region.

Step 1. Calculate $P(X \geq 11) = 1 - P(X \leq 10) = 1 - 0.9574 = 0.0426$. **(M1A1)**

Step 2. Compare to the significance level: $0.0426 < 0.05$. **(M1)**

Answer: Since $0.0426 < 0.05$, the result is significant at the 5% level, so we reject H_0 and conclude there is evidence the mean rate of calls has increased — the same conclusion as before, reached more directly. **(A1)**

EXAM TIP

Both methods (critical region vs p-value) are fully valid and mark schemes accept either, but exam questions sometimes explicitly ask for “the critical region,” in which case the p-value shortcut alone will not earn full marks. Learn both, but read the command word in the question carefully.

Chapter summary

Idea	Key point
Testing framework	State H_0 and H_1 , identify one- or two-tailed, find the critical region, compare, conclude in context
Poisson test	Same logic as Binomial test, using $Po(\lambda_0)$ cumulative probabilities under H_0
One-tailed critical region	Smallest/largest c so that the tail probability is $\leq$ significance level
Two-tailed critical region	Split the significance level equally between both tails
p-value approach	Compare $P(\text{as extreme as observed})$ directly to the significance level
Conclusion	Always written in the context of the original question, never just “reject H_0 ”

Common exam mistakes

- Getting the direction of the critical region backwards relative to H_1 .
- Forgetting to halve the significance level for a two-tailed test.
- Quoting the nominal significance level (e.g. “5%”) as the actual significance level instead of the true tail probability of the critical region found.
- Mixing up $P(X \leq c)$ and $P(X < c)$ (equivalently $P(X \geq c)$ vs $P(X > c)$) when reading cumulative tables — always convert “more than c ” to “ $c+1$ or more” (i.e. $1 - P(X \leq c)$) before looking anything up.
- Concluding “ H_0 is true” rather than the correct “insufficient evidence to reject H_0 ” when the result is not significant.
- Forgetting to write the conclusion in context (in terms of calls, defect rate, etc.) rather than just in symbols.

You should now be able to

- Carry out a full hypothesis test for a Poisson mean, including finding a critical region.
- Carry out a two-tailed hypothesis test for a Binomial proportion or Poisson mean, correctly splitting the significance level.

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- Use either the critical region method or a direct p-value comparison, and know when each is required.
 - State conclusions clearly, in context, using the language of accepting or rejecting H_0 appropriately.